Automated Reasoning and In-Worker Verification for High-Rank Elliptic Curves: The 4-Stage Werr Algebraic Sieve, Conductor Certification ($\log N$), and Vélu Isogeny Graph Transport on the NSF ICARM Leaderboard

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Abstract

Certifying the arithmetic conductor N of high-rank elliptic curves over $\mathbb{Q}$ is a central bottleneck in computational number theory. On the NSF ICARM Elliptic Curve Rank Leaderboard (Grant DMS-2425401), candidate curves are verified in an isolated Cloudflare Worker running WebAssembly PARI/GP 2.17.2. While the kernel certifies point independence via an exact 2-descent quadratic-character certificate over $\mathbb{F}_2$ (`erk_cert`) and computes the minimal model, minimal discriminant $|\Delta_{\min}|$, Faltings height $h_F(E)$, and naive height $h_{\text{naive}}(E)$ via `ellminimalmodel` without factoring $\Delta_{\min}$, automatic conductor verification (`autoBadPrimes`) is bounded by trial division up to 10^7. Whenever $|\Delta_{\min}|$ (50–110 digits) has multiple prime factors above 10^7, unassisted verification returns `conductor: null` (`?`).

We resolve this bottleneck by uniting the *4-Stage Werr Multi-Layer Algebraic Sieve* (Weierstrass covariant gcd kernels, syzygy cubic peeling $\Delta = d_0 s^3$, 2-torsion root-difference splitting, parallel Pollard $\rho/p-1$/ECM, and word-packed $\mathbb{F}_2$ MPQS) with systematic *Vélu Isogeny Graph Transport* ($n \in \{2,3,4,5,6,8,10,12\}$) and 12-case point-interpolation recovery. On the ICARM leaderboard (`user/114`), we established 230 certified Top-10 curves (`#4308`–`#4537`) across 8 torsion groups with 100% certified conductors (230/230), securing 30 #1 Gold records, 80 Top-3 podiums, and 418 Top-10 placements across 43 leaderboards, including Grand Slam `#4315` ($\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, Rank ≥ 9, #1 in all 4 metrics), 110-digit discriminant `#4383` (Rank ≥ 12), and Elkies Rank ≥ 19 curve `#4323` ($\log N = 141.86$; DOI: `10.5281/zenodo.23436282`).

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1 Introduction and Computer-Aided Reasoning in Arithmetic Geometry

The synergy between automated deduction, formal proof certificate checking, and computational number theory has reshaped how mathematical records are established and audited [1, 2, 11]. Historically, computational benchmarks in arithmetic geometry—most notably tables of elliptic curves over $\mathbb{Q}$ with record Mordell–Weil rank or prescribed rational torsion subgroups—were curated as static webpages or text repositories. While foundational to modern number theory, static tables rely on off-line scripts and frequently omit computationally expensive global invariants.

A major advance toward zero-trust, machine-certified mathematical benchmarking is the **NSF Institute for Computer-Aided Reasoning in Mathematics (ICARM)**, established under U.S. National

Science Foundation Grant DMS-2425401 (https://icarm.io). As a flagship benchmark for computer-aided mathematical discovery, ICARM operates the official **Elliptic Curve Rank Leaderboard** (https://elliptic-rank.icarm.cloud), architected and maintained by David Renshaw and Seewoo Lee [12]. Rather than trusting contributor assertions, the ICARM platform executes a deterministic, floating-point-free verification pipeline inside a sandboxed Cloudflare Worker running a precompiled WebAssembly (WASM) binary (`gp-sta.wasm`) of **PARI/GP 2.17.2** [15] with 64 MB of fixed linear memory and a 16 MB PARI stack.

1.1 The ICARM In-Worker Formal Verification Architecture

An examination of the ICARM verification kernel (`src/verify.ts` and `src/pari.ts` in [12]) reveals a remarkably clean automated-reasoning architecture. When an external prover submits an integer Weierstrass 5-tuple $\mathbf{a} = [a_1, a_2, a_3, a_4, a_6] \in \mathbb{Z}^5$, a sequence of n rational points $\mathcal{P} = \{P_1, \ldots, P_n\} \subset \mathbb{Q}^2$, and an optional prime certificate $\mathcal{S} = \{p_1, \ldots, p_k\}$, the in-worker kernel executes six verification stages:

1. **Geometric Non-Singularity and Minimal Model Reduction:** The kernel initializes $E =$ `ellinit`$(\mathbf{a})$, verifies $\Delta(\mathbf{a}) \neq 0$, and computes the unique normalized global minimal model $E_{\min} =$ `ellminimalmodel`$(E, \&v)$ with $a_1, a_3 \in \{0, 1\}$ and $a_2 \in \{-1, 0, 1\}$. Crucially, `ellminimalmodel` determines the canonical isomorphism key `"c4:c6"`, the minimal discriminant $\Delta_{\min} =$ `Emin.disc`, the naive height

$$h_{\text{naive}}(E) = \log \max\big(|c_4(E_{\min})|^3,\, c_6(E_{\min})^2\big)\,, \tag{1}$$

   and the LMFDB-normalized stable Faltings height from the minimal period lattice $\Omega = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2$:

$$h_F(E) = -\frac{1}{2} \log \left|\mathrm{Im}(\overline{\omega_1}\omega_2)\right|\,, \tag{2}$$

   without factoring the discriminant $\Delta_{\min}$.
2. **Exact Rational Point Membership and Transport:** Every submitted coordinate pair $P_j = (x_j, y_j) \in \mathbb{Q}^2$ is verified in exact rational arithmetic via `ellisoncurve(E, P_j)` and mapped onto $E_{\min}(\mathbb{Q})$ via `ellchangepoint(P_j, v)`.
3. **Torsion Subgroup Certification:** Using Lutz–Nagell bounds and division polynomial root analysis via `elltors(Emin)`, the kernel certifies the exact rational torsion structure $E(\mathbb{Q})_{\text{tors}} \cong T$ among the 15 admissible groups of Mazur's Torsion Theorem [14].
4. **Floating-Point-Free 2-Descent Rank Independence Certificate (`erk_cert`):** Rather than relying on floating-point height pairings to certify rank, the ICARM kernel implements the *exact algebraic 2-descent quadratic-character certificate* of Cremona [5] (Section 2, attributed to Brumer) and Silverman [18] (Appendix D). Transforming $E_{\min}$ via $v_6 = [1/6, -b_2/12, -a_1/2, -(a_3 - b_2 a_1/12)/2]$ to the integral short Weierstrass curve

$$F: \quad y^2 = f(x) = x^3 - 27c_4(E_{\min})x - 54c_6(E_{\min}), \qquad \Delta(F) = 6^{12}\Delta_{\min}, \tag{3}$$

   the kernel iterates over good auxiliary primes $p < 10^6$ ($p \nmid 6\Delta_{\min}$) for which $f(x) \bmod p$ has $k \geq 1$ roots $\theta_1, \ldots, \theta_k \in \mathbb{F}_p$. At each such prime, it evaluates the group homomorphism

$$\varepsilon_p : F(\mathbb{Q})/2F(\mathbb{Q}) \longrightarrow (\mathbb{Z}/2\mathbb{Z})^{\min(k,2)}, \qquad \varepsilon_p(P)_j = \frac{1 - \left(\frac{a_j}{p}\right)}{2} \in \mathbb{F}_2, \tag{4}$$

   where for $P = (u/w^2, v/w^3) \in F(\mathbb{Q})$ in lowest terms, $a_j = (u - \theta_j w^2) \bmod p$ if $u \not\equiv \theta_j w^2 \pmod p$, and $a_j = f'(\theta_j) = (3\theta_j^2 - 27c_4) \bmod p$ when $u \equiv \theta_j w^2 \pmod p$. Stacking the ε_p-vectors of the even-order generators of $F(\mathbb{Q})_{\text{tors}}$ into a binary matrix M_{tors} and those of the n candidate points into M_{pts}, the kernel certifies that $P_1, \ldots, P_n$ are $\mathbb{Z}$-linearly independent in $E(\mathbb{Q})/E(\mathbb{Q})_{\text{tors}}$ if and only if

$$\mathrm{rank}_{\mathbb{F}_2}\begin{pmatrix} M_{\text{tors}} \\ M_{\text{pts}} \end{pmatrix} - \mathrm{rank}_{\mathbb{F}_2}(M_{\text{tors}}) = n. \tag{5}$$

   If a kernel combination $Q + \tau \in 2F(\mathbb{Q})$ is divisible by 2 in $F(\mathbb{Q})$ (verified algebraically via `nfroots` on the degree-4 doubling polynomial in `erk_half2`), the kernel replaces the largest-height point in the dependency with its half-point R (up to 40 halvings).
5. **Informational Regulator Diagnostic:** At working precision $\min(250, \max(38, 38{+}2n))$ decimal digits, the kernel evaluates the Néron–Tate regulator $R = \det$ `ellheightmatrix`$(E_{\min}, \mathcal{P})$ purely as an informational diagnostic.

6. **Prime-Gated Conductor Verification (`autoBadPrimes` vs. `conductorFromPrimes`):** Because calling `ellglobalred` triggers unbounded integer factorization inside PARI/GP, the ICARM kernel *never calls `ellglobalred`*. Instead, if no prime certificate $\mathcal{S}$ is supplied, `autoBadPrimes` performs trial division on $|\Delta_{\min}|$ up to `AUTO_FACTOR_BOUND` $= 10^7$ (`factor(abs(E.disc), 10^7)`). It succeeds **only** if every residual cofactor $b > 10^7$ is a Baillie–PSW probable prime (`ispseudoprime(b)`) or a prime power (`ispower(b, , &r) && ispseudoprime(r)`). Otherwise, it immediately aborts and returns `conductor: null` (`?`).

1.2 The Conductor Bottleneck on High-Rank Leaderboards

Across the ICARM platform, curves in each (T, r) category ($T = E(\mathbb{Q})_{\text{tors}}$, rank $\geq r$) are ranked across four official minimality leaderboards:

$$\text{(i) Conductor } N, \quad \text{(ii) Discriminant } |\Delta_{\min}|, \quad \text{(iii) Faltings Height } h_F(E), \quad \text{(iv) Naive Height } h_{\text{naive}}(E). \tag{6}$$

In high-rank constructions pioneered by Noam Elkies [9,10], Andrej Dujella [8], Michael Rubinstein [16], and Zev Klagsbrun [10], curves of rank $r \in [7, 29]$ are obtained by specializing multi-parameter K3 and rational elliptic surfaces at rational parameters of large height. Because $|\Delta_{\min}|$ typically spans **50 to 110 decimal digits** (165 to 365 bits), any discriminant containing two or more prime factors above 10^7 fails `autoBadPrimes`, leaving its conductor marked as `?` and disqualifying the curve from the Conductor N leaderboard. Moreover, many high-rank curves in historical archives are recorded in a single Weierstrass coordinate system tied to their fibration parameterization, leaving the surrounding rational Vélu isogeny graph unexplored.

1.3 Summary of Contributions

In this work, we unite automated proof-certificate synthesis and computational arithmetic geometry through three primary contributions:

1. **The 4-Stage Werr Multi-Layer Algebraic Sieve for Elliptic Discriminants:** We design a four-stage factorization and certificate synthesis architecture (Section 3) that combines Stage 1 Weierstrass covariant gcd kernels ($\gcd(a_4, a_6, \Delta, c_4, c_6)$) and division-polynomial root bracketing, Stage 2 elliptic syzygy cubic divisor peeling ($\Delta = d_0 \cdot s^3$) and 2-torsion pairwise root-difference decomposition, Stage 3 parallel Pollard–Brent $\rho/p - 1$ and multi-curve ECM, and Stage 4 word-packed $\mathbb{F}_2$ MPQS closure to completely factor composite discriminants up to 110 decimal digits [6,7].
2. **Systematic Vélu Isogeny Graph Transport and 12-Case Point-Interpolation Recovery:** We implement an exact rational isogeny graph explorer covering subgroups of orders $n \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, 12\}$ together with a 12-case normalized Weierstrass point-interpolation solver (Section 4) that reconstructs missing Weierstrass models from archival point tables and transports Mordell–Weil generators $P_j \mapsto \phi(P_j)$ across 2- and 3-isogeny graphs.
3. **100% Conductor Certification and 30 #1 Gold Records on NSF ICARM:** We submitted and certified **230 distinct Top-10 curves** (`#4308`–`#4537`) under contributor profile `user/114` (`https://elliptic-rank.icarm.cloud/user/114`), achieving **100.0% conductor certification** (230/230 with zero unknown conductors), **30 #1 Gold records**, **80 Top-3 podiums**, and **418 Top-10 metric placements** across 43 active leaderboards (Section 5).

2 Conductor Determination as Formal Proof Certificate Checking

Let $E/\mathbb{Q}$ be an elliptic curve defined by a generalized Weierstrass equation with integer coefficients:

$$E: \quad y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \quad a_1, a_2, a_3, a_4, a_6 \in \mathbb{Z}. \tag{7}$$

The standard Tate quantities b_2, b_4, b_6, b_8 and covariants c_4, c_6, Δ are given by:

$$b_2 = a_1^2 + 4a_2, \qquad b_4 = 2a_4 + a_1a_3, \qquad b_6 = a_3^2 + 4a_6, \tag{8}$$

$$b_8 = a_1^2a_6 + 4a_2a_6 - a_1a_3a_4 + a_2a_3^2 - a_4^2, \tag{9}$$

$$c_4 = b_2^2 - 24b_4, \qquad c_6 = -b_2^3 + 36b_2b_4 - 216b_6, \tag{10}$$

$$\Delta = -b_2^2b_8 - 8b_4^3 - 27b_6^2 + 9b_2b_4b_6 = \frac{c_4^3 - c_6^2}{1728}. \tag{11}$$

An admissible change of coordinates $(x, y) = (u^2x' + r, u^3y' + u^2sx' + t)$ with $u \in \mathbb{Q}^\times$ and $r, s, t \in \mathbb{Q}$ scales the covariants by $c_4' = u^{-4}c_4$, $c_6' = u^{-6}c_6$, and $\Delta' = u^{-12}\Delta$. Because $\mathbb{Z}$ is a principal ideal domain, $E/\mathbb{Q}$ admits a unique normalized global minimal model $E_{\min}$ satisfying $a_1, a_3 \in \{0, 1\}$, $a_2 \in \{-1, 0, 1\}$, and minimal valuation $v_p(\Delta_{\min})$ simultaneously at all primes p.

Definition 2.1 (Arithmetic Conductor and Ogg–Saito Formula)**.** Let $E_{\min}/\mathbb{Q}$ be the global minimal Weierstrass model with minimal discriminant $\Delta_{\min}$. The *arithmetic conductor* of E is the positive integer

$$N(E) = \prod_{p \mid \Delta_{\min}} p^{f_p}, \qquad f_p = v_p(\Delta_{\min}) - m_p + 1, \tag{12}$$

where m_p is the number of irreducible components (counted without multiplicity) of the special fiber of the Néron minimal model of E over $\mathbb{Z}_p$. Specifically, $f_p = 0$ for good reduction ($p \nmid \Delta_{\min}$, Kodaira type I_0); $f_p = 1$ for multiplicative reduction ($p \mid \Delta_{\min}, p \nmid c_4$, type I_n with $n = v_p(\Delta_{\min}) \geq 1$); and $f_p = 2 + \delta_p \geq 2$ for additive reduction ($p \mid \gcd(c_4, \Delta_{\min})$), where the wild Swan ramification invariant $\delta_p = 0$ for all $p \geq 5$, while $0 \leq \delta_3 \leq 3$ ($f_3 \leq 5$) and $0 \leq \delta_2 \leq 6$ ($f_2 \leq 8$).

Proposition 2.2 (Polynomial-Time Verification of `conductorFromPrimes`)**.** *Let $E_{\min}/\mathbb{Q}$ be the global minimal Weierstrass model computed by* `ellminimalmodel(E)` *with minimal discriminant $\Delta_{\min} \neq 0$. Suppose an external prover supplies a finite set of candidate primes $\mathcal{S} = \{p_1, p_2, \ldots, p_k\} \subset \mathbb{Z}_{\geq 2}$. Then the ICARM verification kernel (*`conductorFromPrimes` *in [12]) certifies the exact global conductor $N(E)$ in deterministic time polynomial in $\log|\Delta_{\min}|$ via the four-step decision procedure:*

1. ***Primality Verification:*** *Verify* $\texttt{ispseudoprime}(p_i) = 1$ *for every $p_i \in \mathcal{S}$;*
2. ***Trial-Division Completeness:*** *Initialize $d \leftarrow |\Delta_{\min}|$, divide out $p_i^{v_{p_i}(d)}$ for each $p_i \in \mathcal{S}$, and verify that the cofactor residual satisfies $d = 1$;*
3. ***Bad-Prime Filtering:*** *Restrict to the sorted subset of primes dividing $\Delta_{\min}$:*

$$\mathcal{B} = \texttt{vecsort}(\{p \in \mathcal{S} : \Delta_{\min} \bmod p = 0\})\,; \tag{13}$$

4. ***Local Tate Reduction Product:*** *For each $p \in \mathcal{B}$, execute Tate's local reduction algorithm [19] via* $\texttt{elllocalred}(E_{\min}, p)[1]$ *to obtain f_p and return $N(E) = \prod_{p \in \mathcal{B}} p^{f_p}$.*

Proof. Step 1 requires k Baillie–PSW / APR-CL primality tests [3], each running in $O((\log p_i)^{4+\varepsilon})$ bit operations. Step 2 performs at most $\sum_{p \in \mathcal{B}} v_p(\Delta_{\min}) \leq \log_2|\Delta_{\min}|$ exact multi-precision divisions of bit-length $B = \lceil \log_2|\Delta_{\min}| \rceil$, costing $O(B^2)$ bit operations, and proves that $\mathcal{B}$ contains every prime divisor of $\Delta_{\min}$. Step 3 cleanly discards any extraneous primes $p \in \mathcal{S} \setminus \mathcal{B}$ (such as primes dividing a non-minimal scaling factor u of the submitted model $\mathbf{a}$ where $\Delta(\mathbf{a}) = u^{12}\Delta_{\min}$). In Step 4, since $E_{\min}$ is already globally minimal, Tate's local algorithm [4, 19] executes a single pass at each $p \in \mathcal{B}$, evaluating p-adic valuations and finding roots over $\mathbb{F}_p$ of auxiliary polynomials of degree ≤ 3 in $O(\log^2 p)$ field operations. Since $|\mathcal{B}| \leq \log_2|\Delta_{\min}|$, the total verification time is $O(\mathrm{poly}(\log|\Delta_{\min}|))$. □

Remark 2.3 (The `cs.LO` Certificate Paradigm: Search vs. Verification). Both the Cremona–Brumer–Silverman 2-descent rank certificate (`erk_cert`, Eq. (5)) and the Tate local reduction conductor certificate (`conductorFromPrimes`, Proposition 2.2) embody the core principle of logic in computer science (`cs.LO`): *replacing intractable or heuristic in-kernel search with a compact, deterministically verifiable algebraic witness.* Our campaign leverages both submission endpoints exposed by the ICARM API: `POST /api/submit` (supplying `ainvs`, `points`, and `primes` simultaneously for new curves) and `POST /api/curve/:id/primes` (retroactively certifying missing conductors on existing leaderboard curves).

3 The 4-Stage Werr Multi-Layer Algebraic Sieve

For high-rank elliptic curves with 50 to 110 digit discriminants, treating $|\Delta|$ as an unstructured black-box integer causes standard factoring routines to stall. To overcome this, we developed the **4-Stage Werr Multi-Layer Algebraic Sieve** [6, 7], which couples elliptic-curve-specific algebraic decompositions (Stages 1–2) with parallel sub-exponential cofactor sieving (Stages 3–4), as summarized in Table 1.

3.1 Stage 1: Root-Kernel Covariant GCDs and Division-Polynomial Decomposition

Before invoking probabilistic or sub-exponential factoring routines, Stage 1 strips all small primes $p \leq 10^5$ via a fast byte-sieve wheel (`SMALL_PRIMES`) and evaluates the mutual greatest common divisors across the Weierstrass coefficients and Tate covariants:

$$g_1 = \gcd(|c_4|, |c_6|), \quad g_2 = \gcd(|a_4|, |a_6|), \quad g_3 = \gcd(|b_2|, |b_4|, |b_6|, |b_8|), \quad g_4 = \gcd(|\Delta|, |c_4|, |c_6|). \tag{14}$$

Whenever $g_k > 1$, Euclidean gcd splitting immediately factors Δ into smaller coprime components Δ/g_k^m and g_k. Furthermore, when E possesses non-trivial rational torsion, the division polynomials $\psi_n(x)$ induce exact polynomial factorizations of Δ:

Table 1: Architecture of the 4-Stage Werr Multi-Layer Algebraic Sieve for Elliptic Curve Discriminants

Pipeline Stage	Algebraic Identity / Method	Target Scale	Mechanism & Role in Conductor Certification
Stage 1 Root-Kernel & GCD Peeling	$\gcd(a_4, a_6, \Delta, c_4, c_6, b_2, b_4, b_6)$ Division polynomials $\psi_2(x), \psi_3(x)$	Smooth towers & algebraic roots	Extracts common polynomial divisor towers across Weierstrass invariants and locates exact integer roots of monic 2-torsion cubics and 3-torsion quartics via derivative-bracketed bisection.
Stage 2 Cubic & Syzygy Divisor Peeling	$c_4^3 - 1728\Delta = c_6^2 \implies \Delta = d_0 \cdot s^3$ $\Delta = 2^{-12}\prod_{i<j}(X_i - X_j)^2$	40–110 digit structured Δ	Isolates cubic divisor towers $s^3 \mid \gcd(c_4^3, \Delta)$, strips non-minimal scaling factors u^{12}, and decomposes 2-torsion discriminants into squares of pairwise root differences $\lvert X_i - X_j\rvert$.
Stage 3 Parallel Pollard ρ, $p-1$ & ECM	Batched Brent ρ (2.5×10^5 steps) Montgomery/Weierstrass ECM ($B_1 = 10^5, B_2 = 5 \times 10^6$)	12–32 digit prime factors	Multi-core worker threads sweep residual composite cofactors in parallel using elliptic curve scalar multiplications modulo C to extract medium-to-large prime factors above 10^7.
Stage 4 Word-Packed $\mathbb{F}_2$ MPQS & APR-CL	Multipolynomial Quadratic Sieve 64-bit word-packed $\mathbb{F}_2$ Gaussian PARI/GP 2.17.2 APR-CL closure	60–105 digit composites (18–38d primes)	Splits remaining balanced composite cofactors via word-packed $\mathbb{F}_2$ nullspace elimination and certifies all prime factors via deterministic APR-CL and Baillie–PSW tests for `conductorFromPrimes`.

Proposition 3.1 (Algebraic Discriminant Decomposition via Rational 2-Torsion)**.** *Let $E/\mathbb{Q}$ be an integral Weierstrass model possessing at least one rational 2-torsion point $P_0 = (x_0, y_0) \in E(\mathbb{Q})[2] \setminus \{O\}$. Then $X_0 = 4x_0 \in \mathbb{Z}$ is an integer root of the monic cubic polynomial*

$$f(X) = X^3 + B_2X^2 + B_4X + B_6 = 0, \quad B_2 = b_2, \quad B_4 = 8b_4, \quad B_6 = 16b_6 \in \mathbb{Z}. \tag{15}$$

Factoring $f(X) = (X - X_0)(X^2 + \alpha X + \beta)$ with $\alpha = B_2 + X_0 \in \mathbb{Z}$ and $\beta = B_4 + \alpha X_0 \in \mathbb{Z}$ decomposes the elliptic discriminant $\Delta(E)$ algebraically into the product:

$$\Delta(E) = 2^{-12} \cdot (\alpha^2 - 4\beta) \cdot (3X_0^2 + 2B_2X_0 + B_4)^2. \tag{16}$$

In particular, if $E(\mathbb{Q})[2] \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ with three integer roots $X_0, X_1, X_2 \in \mathbb{Z}$ of $f(X)$, then $\Delta(E)$ splits completely into the square of the pairwise root differences:

$$\Delta(E) = 2^{-12} \cdot (X_0 - X_1)^2(X_0 - X_2)^2(X_1 - X_2)^2. \tag{17}$$

Proof. A point $P_0 = (x_0, y_0)$ satisfies $2P_0 = O$ if and only if $2y_0 + a_1x_0 + a_3 = 0$. Completing the square in (7) gives $(2y + a_1x + a_3)^2 = 4x^3 + b_2x^2 + 2b_4x + b_6$. Multiplying by 16 and setting $X = 4x$ yields the monic integer cubic $f(X) = X^3 + B_2X^2 + B_4X + B_6$. By Gauss's Lemma, any rational root of $f(X)$ is an integer $X_0 \in \mathbb{Z}$. Since $\mathrm{Disc}(f) = 2^{12}\Delta(E)$, applying the resultant identity $\mathrm{Disc}((X - X_0)q(X)) = \mathrm{Disc}(q) \cdot f'(X_0)^2$ yields (16) and (17). □

Crucially, finding the integer roots $X_0 \in \mathbb{Z}$ of $f(X)$ requires *zero integer factorization*: Stage 1 computes the Cauchy bit-length bound $M = 2^{\max_i(\lceil \mathrm{bits}(|c_i|)/i \rceil)+3}$, brackets the real critical points of $f'(X)$ recursively, and locates all integer roots in $O(\log |B_6|)$ arithmetic operations via exact integer bisection (`integer_roots_monic_cubic`). For curves with full 2-torsion $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ (such as the 110-digit discriminant curve `#4383`), identity (17) immediately reduces the problem of factoring a 110-digit integer $|\Delta|$ to factoring three root-difference integers $|X_i - X_j|$ of at most 18–20 digits each!

3.2 Stage 2: Elliptic Syzygy and Cubic Divisor Peeling

For curves with trivial 2-torsion ($E(\mathbb{Q})[2] = \{O\}$), Stage 2 exploits the fundamental Weierstrass syzygy:

$$c_4^3 - 1728\Delta = c_6^2 \implies c_4^3 \equiv c_6^2 \pmod{1728\Delta}. \tag{18}$$

Any prime $p \geq 5$ dividing both c_4 and Δ (primes of additive reduction) satisfies $p^3 \mid c_4^3$ and $p^2 \mid c_6^2$, allowing Stage 2 to isolate cubic and square divisor towers $\Delta = d_0 \cdot s^3$ via iterated $\gcd(\Delta, c_4^3)$ and exact integer square-root peeling $\mathtt{isqrt}(C)$. Moreover, when a model produced by fibration specialization or Vélu transport is non-minimal with scaling factor $u = 2^\alpha 3^\beta p_1^{\gamma_1} \cdots p_m^{\gamma_m}$, Stage 2 extracts $u^{12} \mid \gcd(c_4^3, c_6^2, \Delta)$ via `approx_minimal_c4_c6_disc`, peeling the 12th-power scaling factor prior to cofactor sieving.

3.3 Stage 3: Parallel Pollard ρ, Pollard $p-1$, and Multi-Curve ECM

Each remaining composite cofactor $C > 1$ (after deterministic Miller–Rabin primality testing across all 12 bases $a \in \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37\}$ and exact square peeling) is pushed onto a factorization work-stack and processed by Stage 3:

1. **Batched Pollard–Brent ρ (`brent_rho`):** Workers execute Brent's cycle-finding improvement to Pollard's ρ algorithm with polynomial transition $x \mapsto (x^2+c) \bmod C$, accumulating modular differences $\prod(x-y) \bmod C$ in batches of 128 steps before computing $\gcd(\text{prod}, C)$ up to 2.5×10^5 iterations. This strips prime factors up to 15–16 decimal digits in milliseconds.
2. **Multi-Curve Weierstrass/Montgomery ECM (`ecm_stage1`):** For cofactors containing 15 to 32 digit primes above ICARM's 10^7 auto-factor bound, Stage 3 launches parallel Lenstra Elliptic Curve Method (ECM) sweeps [13]. Each trial selects random $(x_0, y_0, a) \in (\mathbb{Z}/C\mathbb{Z})^3$, sets $b \equiv y_0^2 - x_0^3 - ax_0 \pmod C$, checks $\gcd(4a^3 + 27b^2, C)$, and multiplies $P_0 = (x_0, y_0)$ by prime powers $p^k \le B_1$ ($B_1 = 10^4$–10^5, Stage 2 continuation $B_2 = 5 \times 10^6$). Whenever an affine chord/tangent denominator ($x_Q - x_R$) or $2y_Q$ is non-invertible modulo C, $\gcd(\text{den}, C) \in (1, C)$ immediately yields a prime factor of C.

3.4 Stage 4: Word-Packed $\mathbb{F}_2$ MPQS and PARI/GP 2.17.2 Closure

When a composite cofactor C of 60 to 130 bits resists ECM (such as balanced products of 18 to 22 digit primes in the 74-digit discriminant of the Rank ≥ 19 curve `#4323`), Stage 4 invokes the Werr 4-Pillar Word-Packed $\mathbb{F}_2$ Multipolynomial Quadratic Sieve (`solve_word_packed_f2_qs` from [7]) and PARI/GP 2.17.2 MPQS/APR-CL closure:

- Polynomials $Q(x) = (Ax + B)^2 - C$ are sieved over the quadratic-residue factor base $\mathcal{F} = \{p \le B_{\max} : (\frac{C}{p}) = 1\}$ using Tonelli–Shanks modular square roots.
- Smooth exponent vectors modulo 2 are packed into 64-bit machine words (`uint64`), and $\mathbb{F}_2$-nullspace dependencies are extracted via bitwise MSB-pivot Gaussian elimination to construct $X^2 \equiv Y^2 \pmod C$, yielding $\gcd(X - Y, C) = p$.
- Every extracted factor is certified prime via deterministic APR-CL / Baillie–PSW tests, producing the complete bad-prime certificate $\mathcal{S} = \{p_1, \ldots, p_k\}$ verified by ICARM's `conductorFromPrimes`.

Table 2 presents representative high-rank discriminant factorizations from our 230 certified curves, illustrating how the 4-stage pipeline resolves both 110-digit multi-factor discriminants and 30-to-38 digit prime factors at ranks $r \in [12, 19]$.

Table 2: Representative High-Rank Discriminant Factorizations Certified by the 4-Stage Werr Sieve

Curve ID	Torsion T	Rank	$\lvert\Delta\rvert$ Dig.	N Dig.	$\log N$	Primes	Largest Prime Factor $p_{\max}$ Dividing Δ	$p_{\max}$ Dig.
#4323	Trivial []	≥ 19	74 d	62 d	141.86	9	5925062485692821329 (3 primes: 18d, 18d, 19d)	19 d
#4395	Trivial []	≥ 15	39 d	38 d	87.49	2	197846159326450166819037158651976203 39	**38** d
#4393	Trivial []	≥ 15	38 d	38 d	86.59	3	549109652144331750922854086151 19	**32** d
#4394	Trivial []	≥ 15	39 d	38 d	87.49	4	840598187908115571800080099	27 d
#4397	Trivial []	≥ 14	34 d	34 d	76.85	3	92383824520523308071859650 3101	30 d
#4466	Trivial []	≥ 13	31 d	31 d	69.19	1	1122482466080847283271138480843 (Prime $\lvert\Delta\rvert = N$)	31 d
#4465	Trivial []	≥ 13	30 d	30 d	68.26	2	2210039735244068846771846868 23	30 d
#4383	$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	≥ 12	**110** d	48 d	110.22	**21**	363119 (21 primes via Stage 1–2 cubic peeling)	6 d
#4321	$\mathbb{Z}/5\mathbb{Z}$	≥ 7	65 d	22 d	50.13	8	115925634521	12 d
#4320	$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$	≥ 6	52 d	18 d	39.78	11	947	3 d
#4319	$\mathbb{Z}/4\mathbb{Z}$	≥ 8	50 d	21 d	48.33	11	6367	4 d
#4315	$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	≥ 9	46 d	22 d	50.40	10	1433 (Grand Slam: #1 in all 4 metrics)	4 d

4 Vélu Rational Isogeny Graph Transport and Minimal Model Search

While the 4-Stage Werr Algebraic Sieve resolves the conductor certification bottleneck ($\log N$), our second engine exploits the geometry of rational isogeny graphs to discover strictly smaller minimal Weierstrass models and populate adjacent torsion leaderboards.

4.1 Invariance of Rank and Conductor Across $\mathbb{Q}$-Isogeny Orbits

Let $E/\mathbb{Q}$ and $E'/\mathbb{Q}$ be elliptic curves connected by a non-constant $\mathbb{Q}$-rational isogeny $\phi : E \to E'$ of degree $n = |\ker \phi|$, with dual isogeny $\hat{\phi} : E' \to E$ satisfying $\hat{\phi} \circ \phi = [n]_E$ and $\phi \circ \hat{\phi} = [n]_{E'}$.

Theorem 4.1 (Invariance of Rank, Conductor, and Bad Primes under $\mathbb{Q}$-Isogeny)**.** *Let $\phi : E \to E'$ be a $\mathbb{Q}$-rational isogeny between elliptic curves over $\mathbb{Q}$. Then:*

1. ***Mordell–Weil Rank Invariance:*** *ϕ induces a $\mathbb{Q}$-linear isomorphism $E(\mathbb{Q}) \otimes_{\mathbb{Z}} \mathbb{Q} \xrightarrow{\sim} E'(\mathbb{Q}) \otimes_{\mathbb{Z}} \mathbb{Q}$, so* $\text{rank}(E'(\mathbb{Q})) = \text{rank}(E(\mathbb{Q}))$.

2. ***Conductor and Bad-Prime Certificate Reuse:*** *By the Néron–Ogg–Shafarevich criterion and Faltings' isogeny theorem, $V_\ell(E) \cong V_\ell(E')$ as $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-modules. Consequently, E and E' have the exact same set of bad primes $\mathcal{B}$, the same global arithmetic conductor $N(E') = N(E)$, and identical L-functions $L(E', s) = L(E, s)$. In particular, any prime certificate $\mathcal{S}$ valid for E (augmented by $\{2, 3\}$ for scaling factors) immediately certifies $N(E')$ via* `conductorFromPrimes!`

Crucially, however, the *minimal discriminant* $|\Delta_{\min}(E')|$, the *Faltings height* $h_F(E')$, the *naive height* $h_{\mathrm{naive}}(E')$, and the *rational torsion subgroup* $E'(\mathbb{Q})_{\mathrm{tors}}$ are **not** invariant under isogeny. Under a prime ℓ-isogeny $\phi : E \to E'$, a multiplicative fiber of Kodaira type I_m at $p \mid N$ transitions either to $\mathrm{I}_{m\cdot\ell}$ or to $\mathrm{I}_{m/\ell}$, multiplying or dividing $|\Delta_{\min}|$ by ℓ^m across multiple primes while shifting the real period lattice covolume and Faltings height.

4.2 Explicit Vélu Quotient Equations for n-Isogenies and Pure-$\mathbb{Q}$ 3-Isogenies

Let $G \subset E(\overline{\mathbb{Q}})$ be a finite $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-stable subgroup of order n. Following Vélu [20], we partition $G \setminus \{O\} = G_2 \sqcup G_+ \sqcup G_-$, where $G_2 = \{Q \in G \setminus \{O\} : 2Q = O\}$ and each inverse pair $\{Q, -Q\}$ of order > 2 has one representative in G_+ and one in G_-. Setting $S = G_2 \sqcup G_+$, for each $Q = (x_Q, y_Q) \in S$ our engine (`werr_elliptic_rank_engine.py`) computes in exact rational arithmetic (`fractions.Fraction`):

$$g_Q^x = 3x_Q^2 + 2a_2x_Q + a_4 - a_1y_Q, \qquad g_Q^y = -2y_Q - a_1x_Q - a_3, \tag{19}$$

$$t_Q = \begin{cases} g_Q^x & \text{if } Q \in G_2, \\ 2g_Q^x - a_1g_Q^y & \text{if } Q \in G_+, \end{cases} \qquad u_Q = (g_Q^y)^2, \qquad w_Q = u_Q + x_Qt_Q. \tag{20}$$

Summing $t = \sum_{Q\in S} t_Q$ and $w = \sum_{Q\in S} w_Q$, the quotient curve $E' = E/G$ is given by $\mathbf{A} = [a_1, a_2, a_3, a_4 - 5t, a_6 - b_2t - 7w]$, and every independent Mordell–Weil generator $P_j = (x_j, y_j) \in \mathcal{P}$ is transported via the exact rational map $\phi(P_j) = (X_j, Y_j) \in E'(\mathbb{Q})$:

$$X_j = x_j + \sum_{Q\in G\setminus\{O\}} \left(x(P_j + Q) - x(Q)\right), \qquad Y_j = y_j + \sum_{Q\in G\setminus\{O\}} \left(y(P_j + Q) - y(Q)\right). \tag{21}$$

For rational 3-isogenies (`submit_mu3_breakthroughs.py`), a $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-stable subgroup $G = \{O, Q, -Q\}$ of order 3 corresponds to a rational root $x_0 = X_0/3 \in \mathbb{Q}$ of the 3-division polynomial $\psi_3(x) = 3x^4 + b_2x^3 + 3b_4x^2 + 3b_6x + b_8 = 0$, where $X_0 \in \mathbb{Z}$ is an integer root of the monic quartic

$$P_4(X) = X^4 + b_2X^3 + 9b_4X^2 + 27b_6X + 27b_8 = 0. \tag{22}$$

Observe that $t_Q = 2g_Q^x - a_1g_Q^y = 6x_0^2 + b_2x_0 + b_4$ and $u_Q = (g_Q^y)^2 = 4x_0^3 + b_2x_0^2 + 2b_4x_0 + b_6$ depend *only on* $x_0 \in \mathbb{Q}$ and not on $y_0 \in \mathbb{Q}(\sqrt{u_Q})$! Consequently, our function `velu_3isogeny_from_x0` evaluates both E/G and the transported generator coordinates purely over $\mathbb{Q}$:

$$\begin{aligned} X_j &= x_j + \frac{t_Q}{x_j - x_0} + \frac{u_Q}{(x_j - x_0)^2}, \\ Y_j &= \frac{1}{2}\left((2y_j + a_1x_j + a_3)\left(1 - \frac{t_Q}{(x_j - x_0)^2} - \frac{2u_Q}{(x_j - x_0)^3}\right) - a_1X_j - a_3\right). \end{aligned} \tag{23}$$

Furthermore, if $u_Q \notin (\mathbb{Q}^\times)^2$ but $-3u_Q \in (\mathbb{Q}^\times)^2$, then $G \cong \mu_3$ has trivial rational 3-torsion on $E(\mathbb{Q})$, whereas the dual kernel $\hat{G} \subset E'(\mathbb{Q})[3]$ is generated by a *rational* point of order 3, transporting a trivial-torsion curve directly onto the $T = \mathbb{Z}/3\mathbb{Z}$ leaderboard!

4.3 12-Case Normalized Weierstrass Point-Interpolation Recovery

In our campaign across 30 archival subpages of Dujella's database [8] and 29 markdown tables of Rubinstein's repository [16], numerous high-rank records (such as `z2z2old*.html` and `z6old*.html`) list the independent generator points $\{P_1, \ldots, P_r\}$ without explicit 5-tuple Weierstrass coefficients $[a_1, a_2, a_3, a_4, a_6]$. Our finalizer engine (`recover_ainvs_from_points_and_text` in `master_werr_campaign_finalizer.py`) reconstructs the exact Weierstrass model via a two-tier exact solver over $\mathbb{Q}$:

1. **Tier 1 (12-Case Normalized Minimal Scan):** Since every global minimal model over $\mathbb{Q}$ satisfies $(a_1, a_2, a_3) \in \{0, 1\} \times \{-1, 0, 1\} \times \{0, 1\}$, for each of the $2 \times 3 \times 2 = 12$ normalized triples and two points P_1, P_2 with $x_1 \neq x_2$, the engine solves the exact 2×2 linear system

$$a_4 = \frac{C(P_1) - C(P_2)}{x_1 - x_2}, \quad a_6 = C(P_1) - a_4x_1, \quad C(x, y) := y^2 + a_1xy + a_3y - x^3 - a_2x^2, \tag{24}$$

and verifies whether $a_4, a_6 \in \mathbb{Z}$ and `on_curve`$(\mathbf{a}, P_j)$ holds for all $1 \leq j \leq r$.

2. **Tier 2 (Full 5×5 Exact Rational Gaussian Elimination):** If the archival points are given on a non-normalized coordinate model (when $r \geq 5$), the engine solves the 5×5 linear system $y_j x_j a_1 - x_j^2 a_2 + y_j a_3 - x_j a_4 - a_6 = x_j^3 - y_j^2$ in `fractions.Fraction` and scales to integers via `scale_to_integer_ainvs`.

Table 3 lists the 8 certified Top-10 curves in our dataset obtained explicitly via Vélu rational 2-isogeny transport across torsion classes ($\mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, $\mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/6\mathbb{Z}$, and $\mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$).

Table 3: Certified Top-10 Leaderboard Curves Obtained via Explicit Vélu Rational 2-Isogeny Transport

Curve ID	Target T	Rank	Source Family & Torsion Transition	$\log N$	$\lvert\Delta\rvert$ Dig.	$h_F(E)$	h_{naive}	Minimal Model $[a_1, a_2, a_3, a_4, \dots]$
#4309	$\mathbb{Z}/2\mathbb{Z}$	≥ 9	2-Isog. of `2-9.ec` ($\mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$)	47.13	32 d	4.75	80.18	$[1, 1, 0, 8425949699, \dots]$
#4385	$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	≥ 9	2-Isog. of `family_Z4#49` ($\mathbb{Z}/4\mathbb{Z} \to [2, 2]$)	64.14	66 d	11.80	170.32	$[0, 0, 0, -943516387852 \dots]$
#4310	$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	≥ 8	2-Isog. of `4-8.ec` ($\mathbb{Z}/4\mathbb{Z} \to [2, 2]$)	48.33	52 d	8.59	128.31	$[1, 0, 0, -781408640210 \dots]$
#4387	$\mathbb{Z}/6\mathbb{Z}$	≥ 8	2-Isog. of `z6old678#123` ($\mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/6\mathbb{Z}$)	60.19	78 d	13.64	186.41	$[0, 1, 0, 166901229740 \dots]$
#4390	$\mathbb{Z}/6\mathbb{Z}$	≥ 8	2-Isog. of `z6old678#124` ($\mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/6\mathbb{Z}$)	61.45	86 d	15.48	213.72	$[1, -1, 0, -18132936152 \dots]$
#4308	$\mathbb{Z}/2\mathbb{Z}$	≥ 7	2-Isog. of `2-7.ec` ($\mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$)	33.04	22 d	3.08	63.24	$[1, -1, 1, -29735621, \dots]$
#4359	$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	≥ 5	2-Isog. of `torsion_Z4:2354` ($\mathbb{Z}/4\mathbb{Z} \to [2, 2]$)	27.04	24 d	4.19	81.12	$[0, 0, 0, -11542418356, \dots]$
#4379	$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$	≥ 4	2-Isog. of `torsion_Z4:352` ($\mathbb{Z}/4\mathbb{Z} \to [4, 2]$)	23.94	32 d	5.21	90.74	$[0, -1, 0, -284743456964, \dots]$

5 Empirical Leaderboard Results on NSF ICARM

Executing the combined 4-Stage Werr Algebraic Sieve and Vélu Isogeny Graph Engine across the live ICARM database and historical parametric curve families, we submitted and certified **230 distinct Top-10 curves** (`#4308` through `#4537`) on `https://elliptic-rank.icarm.cloud` under contributor profile **Volkan Dagli** (`user/114`).

5.1 Global Certification Metrics and Torsion–Rank Distribution

Table 4 summarizes the overall verification audit of our 230 accepted curves certified by the ICARM in-worker PARI/GP 2.17.2 kernel, and Table 5 gives the exact distribution of these 230 curves across all 8 torsion subgroups and 13 rank thresholds ($r = 4$ to $r = 19$).

Table 4: Official Verification and Leaderboard Audit Summary on NSF ICARM (`user/114`)

Leaderboard Metric / Verification Attribute	Certified Value
Total Curves Accepted in Top-10 (`#4308`–`#4537`)	**230 curves** (100.0% `status: created`)
Certified Conductor Factorizations ($\log N$)	**230 / 230** (100.0% **factored, 0 unknown ?**)
Active (T, Rank) Leaderboards with Top-10 Presence	**43 distinct leaderboards**
Total **#1 Gold** (First-Place World Record) Placements	**30 First-Place Records**
Total **Top-3** (Podium: #1, #2, #3) Placements	**80 Podium Finishes**
Total **Top-10** Placements Across All 4 Metrics	**418 Top-10 Placements**
Mordell–Weil Rank Range of Certified Curves	**$r \geq 4$ up to $r \geq 19$**
Largest Minimal Discriminant Completely Factored	**110 decimal digits** (21 distinct prime factors on `#4383`)
Largest Individual Prime Factor Certified in Δ	**38 decimal digits** ($p =$ `19784615932645016681903715865197620339`)

Table 5: Distribution of the 230 Certified Curves Across Torsion Subgroups and Rank Bounds

Torsion T	ICARM Key	r=4	5	6	7	8	9	10	11	12	13	14	15	19	Total
Trivial $\{O\}$	`[]`	–	–	19	19	16	14	2	8	12	12	1	4	1	**108**
$\mathbb{Z}/3\mathbb{Z}$	`[3]`	–	9	8	9	2	–	5	2	–	–	–	–	–	**35**
$\mathbb{Z}/2\mathbb{Z}$	`[2]`	2	3	–	5	6	5	3	3	–	–	–	–	–	**27**
$\mathbb{Z}/5\mathbb{Z}$	`[5]`	12	7	–	1	–	–	–	–	–	–	–	–	–	**20**
$\mathbb{Z}/4\mathbb{Z}$	`[4]`	6	6	3	1	1	–	–	–	–	–	–	–	–	**17**
$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	`[2, 2]`	–	4	7	1	2	1	–	–	1	–	–	–	–	**16**
$\mathbb{Z}/6\mathbb{Z}$	`[6]`	–	1	1	1	1	–	–	–	–	–	–	–	–	**4**
$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$	`[4, 2]`	1	1	1	–	–	–	–	–	–	–	–	–	–	**3**
Total Curves		**21**	**31**	**39**	**37**	**28**	**20**	**10**	**13**	**13**	**12**	**1**	**4**	**1**	**230**

5.2 Complete Breakdown of #1 Gold Records (30 First-Place Placements)

Table 6 presents the 15 flagship curves from our Zenodo monograph [6] that achieved `#1` (First Place / Gold Record) and world-record placements across the four official leaderboard metrics: Conductor (N), Minimal

Discriminant ($|\Delta_{\min}|$), Faltings Height (h_F), and Naive Height (h_{naive}).

Table 6: Summary of #1 Gold Records and Landmark Placements on the NSF ICARM Leaderboard

Torsion T	**Rank**	**Curve ID**	**#1 Gold Record Metrics**	**Additional Podiums & Certified Invariants**
$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	≥ 9	**#4315**	**#1 Conductor (N), #1 $\lvert\Delta\rvert$, #1 Faltings, #1 Naive**	*Grand Slam: 4/4 #1 records* ($\log N = 50.40, h_F = 7.51$)
$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	≥ 8	**#4314**	**#1 Naive Height**	#2 $\lvert\Delta\rvert$, #2 Faltings, #3 N ($\log N = 44.23, h_F = 6.18$)
$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	≥ 7	**#4313**	**#1 $\lvert\Delta\rvert$, #1 Faltings, #1 Naive**	#6 Conductor N ($\log N = 37.00, h_F = 4.90$)
$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$	≥ 6	**#4320**	**#1 $\lvert\Delta\rvert$, #1 Faltings, #1 Naive**	#5 Conductor N ($\log N = 39.78, h_F = 8.68$)
$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$	≥ 5	**#4322**	**#1 $\lvert\Delta\rvert$, #1 Faltings, #1 Naive**	#2 Conductor N ($\log N = 27.61, h_F = 5.74$)
$\mathbb{Z}/4\mathbb{Z}$	≥ 8	**#4319**	**#1 $\lvert\Delta\rvert$, #1 Faltings, #1 Naive**	#3 Conductor N ($\log N = 48.33, h_F = 8.25$)
$\mathbb{Z}/4\mathbb{Z}$	≥ 7	**#4318**	**#1 $\lvert\Delta\rvert$, #1 Faltings, #1 Naive**	#3 Conductor N ($\log N = 40.16, h_F = 6.40$)
$\mathbb{Z}/5\mathbb{Z}$	≥ 7	**#4321**	**#1 $\lvert\Delta\rvert$, #1 Faltings, #1 Naive**	#2 Conductor N ($\log N = 50.13, h_F = 11.15$)
$\mathbb{Z}/5\mathbb{Z}$	≥ 5	**#4380**	**#1 Discriminant $\lvert\Delta\rvert$**	#2 $\lvert\Delta\rvert$ (`#4381`), #3 Faltings ($\log N = 32.61$)
$\mathbb{Z}/2\mathbb{Z}$	≥ 9	**#4312**	**#1 Faltings, #1 Naive Height**	#3 Discriminant $\lvert\Delta\rvert$ ($\log N = 47.13, h_F = 4.41$)
$\mathbb{Z}/2\mathbb{Z}$	≥ 7	**#4311**	**#1 Naive Height**	#2 $\lvert\Delta\rvert$, #2 Faltings (`#4311`), #3 $\lvert\Delta\rvert$ (`#4511`)
$\mathbb{Z}/3\mathbb{Z}$	≥ 8	**#4317**	**#1 Naive Height**	#2 Faltings (`#4317`), #3 $\lvert\Delta\rvert$ (`#4324`)
$\mathbb{Z}/3\mathbb{Z}$	≥ 7	**#4316**	**#1 Faltings, #1 Naive Height**	#2 $\lvert\Delta\rvert$ (`#4327`), #2 Naive (`#4330`)
Trivial `[]`	≥ 19	**#4323**	**Top-10 Overall** (Rank ≥ 19)	#4 Naive, #6 Faltings, #8 $\lvert\Delta\rvert$ ($\log N = 141.86$)
$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	≥ 12	**#4383**	**Highest Rank in $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$**	#2 $\lvert\Delta\rvert$, #2 Faltings, #2 Naive, #5 N ($\log N = 110.22$)

5.3 Detailed Case Studies of Landmark Curves

5.3.1 Case Study 1: The "Grand Slam" Curve (`#4315`, $T = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, Rank ≥ 9)

Curve `#4315` (https://elliptic-rank.icarm.cloud/curve/4315), originating from the Layr-Labs / Yukon Elliptic Rank Challenge (`2x2-9.ec`) and certified with its complete conductor factorization via our 4-Stage Werr Algebraic Sieve, is defined by the minimal Weierstrass equation:

$$E_{4315}: \quad y^2 = x^3 - x^2 - 567553808246049x + 2655925051270455158049. \tag{25}$$

Its exact machine-verified arithmetic invariants and real-locus geometry are:

- **Mordell–Weil Rank & Regulator:** $\operatorname{rank}(E_{4315}(\mathbb{Q})) \geq 9$, certified in 0 halvings via `erk_cert`, with Néron–Tate regulator $R = 3980874.3395$.
- **Real-Locus Topology $E_{4315}(\mathbb{R})$:** Because $E_{4315}(\mathbb{Q})[2] \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, the cubic right-hand side has three rational roots $e_1 < e_2 < e_3$, decomposing $E_{4315}(\mathbb{R})$ into two connected components ($c_\infty = 2$): 7 independent generators $(P_2, \ldots, P_8)$ lie on the compact oval $[-25.89 \times 10^6, 4.88 \times 10^6]$ and 2 generators (P_1, P_9) lie on the unbounded branch $x \geq 21.00 \times 10^6$.
- **Minimal Discriminant (46 digits):** $\Delta_{4315} = 865310980846188352998855768998052485931525734 4$.
- **Certified Bad Primes (10 primes):** $\mathcal{S}_{4315} = \{2, 3, 41, 73, 79, 311, 337, 953, 1193, 1433\}$.
- **Certified Conductor (22 digits):** $N_{4315} = 77518410933494281637 76$ ($\log N = 50.402$, $h_F = 7.5110$, $h_{\text{naive}} = 113.5307$).
- **Leaderboard Outcome:** Captures a **Grand Slam**—ranking **#1 in Conductor N**, **#1 in Discriminant $|\Delta|$**, **#1 in Faltings Height**, and **#1 in Naive Height** simultaneously among all known curves with $T = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ and rank ≥ 9.

5.3.2 Case Study 2: 110-Digit Discriminant Factorization (`#4383`, $T = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, Rank ≥ 12)

Curve `#4383` (https://elliptic-rank.icarm.cloud/curve/4383), discovered by Zev Klagsbrun and cataloged in Dujella's historical tables (`z2z2old1011121314.html#pre19`) at the world-record rank ≥ 12 for $T = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, was reconstructed from its 12 rational generators via our Tier-1 normalized point-interpolation solver to yield the global minimal model (with 36-digit a_4 and 54-digit a_6):

$$\begin{aligned} E_{4383}: \quad y^2 = x^3 - x^2 &- 6773066279657575296262579351595404 33x \\ &+ 1193016672543994534753764971576784359254622489500827 37. \end{aligned} \tag{26}$$

On the real locus $E_{4383}(\mathbb{R})$, 7 independent generators lie on the compact oval and 5 generators lie on the unbounded branch. Its minimal discriminant Δ_{4383} spans **110 decimal digits** (363 bits):

$$\begin{aligned} \Delta_{4383} = 13736866607123642983650570238897861445683593961485813 91 \\ \times 10^{55} + 52602222878509559524517475572750320441108924951040000 00. \end{aligned}$$

Using Proposition 3.1 (Stage 1–2 monic cubic pairwise root-difference decomposition) followed by Stage 3 sieving, our pipeline factored Δ_{4383} completely into **21 distinct prime factors**:

$$\begin{aligned}\mathcal{S}_{4383} = \{&2, 3, 5, 11, 17, 31, 37, 43, 47, 71, 89, 97, 137, 197, 433,\\ &877, 1093, 1153, 2423, 50671, 363119\},\end{aligned} \tag{27}$$

certifying its 48-digit conductor $N_{4383} = 734375670289085539129194693801768623503331947200$ ($\log N = 110.219$, regulator $R = 5.2391 \times 10^{11}$, $h_F = 19.6397$, $h_{\text{naive}} = 259.1239$) and securing `#2` in $|\Delta_{\min}|$, h_F, and h_{naive} on the $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, Rank ≥ 12 leaderboard.

5.3.3 Case Study 3: Elkies Rank ≥ 19 Conductor Certification (`#4323`, Trivial Torsion)

Curve `#4323` (`https://elliptic-rank.icarm.cloud/curve/4323`), specialized by Noam Elkies and cataloged in Rubinstein's tables (`family_trivial.tsv.gz:114`), achieves certified Mordell–Weil rank $\text{rank}(E(\mathbb{Q})) \geq 19$ (regulator $R = 1.8090 \times 10^{21}$) with minimal equation:

$$E_{4323}: \quad y^2 = x^3 - 7394310748527678442369 75x + 331314067119134775943540728349251250. \tag{28}$$

Its 74-digit negative minimal discriminant

$$\Delta_{4323} = -215456471414460843586353053040389451042841241455353311473295739014360000 00$$

fails ICARM's `autoBadPrimes` (10^7 bound) because after stripping small prime powers $2^6 \cdot 5^6 \cdot 7^2 \cdot 13^2 \cdot 47 \cdot 127$, there remains a 54-digit composite cofactor C_{54} composed of three primes of 18, 18, and 19 digits. The 4-Stage Werr Algebraic Sieve resolved all **9 bad primes**:

$$\mathcal{S}_{4323} = \{2, 5, 7, 13, 47, 127, \underbrace{160113632370089479}_{\text{18-digit prime}}, \underbrace{392731995558810161}_{\text{18-digit prime}}, \underbrace{5925062485692821329}_{\text{19-digit prime}}\}, \tag{29}$$

certifying its 62-digit minimal conductor:

$$N_{4323} = 40475339116507435307125435712606155771861961364063680791605800$$

($\log N = 141.860$, $\log|\Delta_{\min}| = 168.862$, $h_F = 12.7770$, $h_{\text{naive}} = 177.0999$), placing `#4323` at `#4` in Naive Height, `#6` in Faltings Height, and `#8` in Discriminant across all elliptic curves of rank ≥ 19.

6 Scientific Impact: Resolving the $\log N$ Upper-Bound Gap

A central problem in analytic and algebraic number theory—investigated by Goldfeld, Mestre, Brumer, Watkins, and Elkies [4, 9, 17]—is determining explicit upper bounds for $\text{rank}(E(\mathbb{Q}))$ in terms of $\log N(E)$. Under the Generalized Riemann Hypothesis (GRH) and the Birch and Swinnerton-Dyer (BSD) conjecture, Mestre's explicit formula bounds the analytic rank by $r = O(\log N / \log\log N)$, while Szpiro's conjecture posits that for any $\varepsilon > 0$ there exists a constant $C(\varepsilon)$ such that $|\Delta_{\min}(E)| \leq C(\varepsilon) \cdot N(E)^{6+\varepsilon}$. Empirical testing of Mestre–Brumer conductor bounds and Szpiro ratios $\sigma(E) = \log|\Delta_{\min}(E)| / \log N(E)$ at extreme ranks ($r \in [8, 19]$) has historically been hampered by the "unfactored conductor (`?`)" gap whenever $|\Delta_{\min}|$ contains multiple primes above 10^7.

By delivering **100% certified prime factorizations** (230/230) across all 230 high-rank curves in our dataset, this work provides a machine-checked empirical benchmark for arithmetic geometers:

- On the $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, Rank ≥ 9 Grand Slam curve `#4315`, the Szpiro ratio is $\sigma(E_{4315}) = \log|\Delta_{\min}| / \log N = 105.78/50.40 = 2.099$, and the rank-to-conductor ratio is $r / \log_{10} N = 9/21.89 = 0.411$.
- On the 110-digit discriminant curve `#4383` ($\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, Rank ≥ 12), Stage 1–2 cubic peeling reveals that heavy prime powers at small primes drop the 110-digit discriminant ($|\Delta_{\min}| \approx 1.37 \times 10^{109}$) to a 48-digit conductor ($N \approx 7.34 \times 10^{47}$), yielding $\sigma(E_{4383}) = 251.30/110.22 = 2.280$.
- On the Elkies Rank ≥ 19 curve `#4323`, the certified conductor $\log N = 141.86$ ($\log_{10} N = 61.61$) establishes an explicit certified high-rank benchmark point with Szpiro ratio $\sigma(E_{4323}) = 168.86/141.86 = 1.190$.

7 Intellectual Attribution and Acknowledgments

In open science and computational mathematics, new algorithmic syntheses and verification campaigns build directly upon decades of foundational theoretical and computational discoveries:

1. **Original Discoverers of High-Rank Parametric Families:** Full credit for the underlying elliptic surface fibrations, Mordell–Weil lattice specializations, and p-adic point searches belongs to their original discoverers: **Prof. Noam D. Elkies** (Harvard University) [9,10], **Prof. Andrej Dujella** (University of Zagreb) [8], **Prof. Michael Rubinstein** (University of Waterloo) [16], **Dr. Zev Klagsbrun** [10], and the contributors of the **Layr-Labs / Yukon Elliptic Rank Challenge**, alongside the foundational methodologies of J. Silverman [17,18], J. Cremona [4,5], K. Eriksson, and A. Brumer.
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8 Conclusion and Future Work

By uniting the 4-Stage Werr Multi-Layer Algebraic Sieve with systematic Vélu Isogeny Graph Transport, we resolved the conductor certification bottleneck across 230 high-rank elliptic curves over $\mathbb{Q}$ on the NSF ICARM leaderboard, achieving 100% certified conductor factorizations (230/230), 30 #1 Gold records, 80 podiums, and 418 Top-10 placements across 43 active leaderboards. Future work will extend this certificate-driven methodology to genus-2 hyperelliptic Jacobians over $\mathbb{Q}$ and formalize both the 2-descent quadratic-character independence certificate (`erk_cert`) and Tate's local reduction algorithm directly in the Lean 4 theorem prover.

Data Availability and Reproducibility

All 230 certified curves, complete Weierstrass models, rational generator coordinates, certified bad prime factorizations, and standalone Python reproduction scripts (`werr_algebraic_sieve.py`, `velu_isogeny_engine.py`, and `verify_leaderboard_audit.py`) are permanently available under open licenses:

- **CERN Zenodo Concept DOI:** `https://doi.org/10.5281/zenodo.23436282`
- **CERN Zenodo Release v1.0 DOI:** `https://doi.org/10.5281/zenodo.23436283`
- **GitHub Repository:** `https://github.com/pCwOrM/werr-elliptic-sieve`
- **Official ICARM Contributor Profile:** `https://elliptic-rank.icarm.cloud/user/114`

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